Content Studio

AI-Powered Multimodal Content Creation Tool on AMD Radeon GPU + ROCm

2026 AMD AI DevMaster · Track 1

1. Project Background

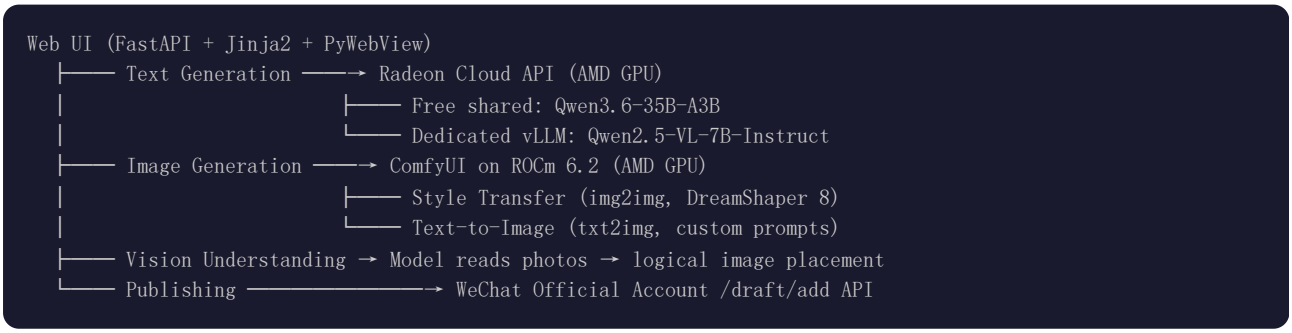
Content creators on Chinese social media (WeChat Official Accounts, Xiaohongshu/RedNote) face a repetitive, multi-step workflow: write articles, style photos, place images, format per platform, and manually publish. No existing tool integrates AI text generation, image styling, intelligent placement, and direct publishing into one pipeline — especially one powered by AMD Radeon GPU + ROCm.

Content Studio fills this gap. Users upload photos, type a theme, and get a complete article with AI-styled images logically placed — all on AMD GPU infrastructure.

2. Target Users & Use Cases

User	Scenario
Social media operators	Daily batch content production for WeChat / Xiaohongshu
Independent creators	Turn personal photos + ideas into polished articles in minutes
Small business marketing	Generate promotional content from product photos
Content teams	Pre-generate drafts for review, then publish via API

3. System Architecture



Key design: `core/` is UI-independent (92 tests). All services communicate via HTTP — mockable, replaceable. Switching LLM models = change config, not code.

4. Model & Algorithm

4.1 Text Generation

- **LLM:** Qwen3.6-35B-A3B (Radeon Cloud) — 1000+ word articles with structured JSON output
- **Robust parsing:** Code-fence stripping, prose extraction, repair re-request, raw fallback
- **Templates:** Product review, event promo, tutorial, industry analysis

4.2 Image Style Transfer (img2img)

- **Engine:** ComfyUI + DreamShaper 8 on AMD GPU (ROCm 6.2, gfx1100)
- **Workflow:** LoadImage → VAEEncode → KSampler (denoise 0.55, 20 steps) → VAEDecode
- **8 presets:** anime, cyberpunk, watercolor, retro, minimal, oil, ink wash, pixel art

4.3 Text-to-Image (txt2img)

- **Engine:** ComfyUI + DreamShaper 8, EmptyLatentImage 512×512, 22 steps
- **Custom prompts:** User writes English descriptions (one per line), ComfyUI renders each

4.4 Vision-Based Image Placement

- **Model:** Qwen3.6-35B reads each photo's content → places `[[IMG:k]]` markers at relevant sections
- **Fallback:** Converts Chinese "图N" references; distributes unplaced images after section headers

4.5 WeChat Draft Publishing

- access_token cached (auto-refresh); body images via `/media/uploadimg`; cover via `/material/add_material`; draft via `/draft/add`
- 40001 token expiry → invalidate + retry once

5. AMD Radeon GPU / ROCm Adaptation

Component	AMD GPU	ROCm Stack
Text generation	Qwen3.6-35B (Radeon Cloud)	Radeon Cloud managed
Dedicated LLM	Qwen2.5-VL-7B via vLLM	ROCm 6.2, PyTorch 2.5.1+rocm6.2
Style transfer	DreamShaper 8 via ComfyUI	ROCm 6.2, gfx1100 (RDNA3)
Vision understanding	Model reads photos for placement	Same as text generation

Hardware verified: AMD Radeon Graphics, gfx1100, 49,136 MB VRAM, ROCm 6.2

Performance Data

Operation	Time
Text generation (free API)	20–300s (shared, variable)
Text generation (dedicated vLLM)	2–30s
Image style transfer (img2img)	5–15s/image
Text-to-image (txt2img)	5–10s/image
App startup	~1.4s (deferred imports)

6. Innovation Highlights

1. **Vision-driven image placement:** Model reads photo content → places each image beside the most relevant paragraph
2. **Custom t2i prompts:** User specifies what ComfyUI should draw — full creative control
3. **Auto-detect vision capability:** Falls back to text-only if model doesn't support images
4. **Direct API publishing:** One click → complete article in WeChat draft box
5. **Dual-platform output:** WeChat articles + Xiaohongshu posts from same input
6. **Graceful degradation:** ComfyUI down → original photos; LLM slow → clear status with elapsed timer

7. What's Included

- 92 automated tests (pytest)
- Complete source code (FastAPI + ComfyUI + WeChat API)
- Desktop application (PyWebView + PyInstaller .exe)
- ROCm migration guide (3 deployment paths)
- English UI + README + project description
- Demo video script (bilingual CN/EN)
- Self-diagnostic page + GPU live status